Assignment 3

Please answer the following. Answer on a physical copy, whole yellow paper or bond paper (can be more than one paper), any black or blue pen, as long as it is **readable**. Label your answers and section them in terms of struct, array, and linked list answers.

For coding questions, as much as possible, try not to use compilers and try to compile it by yourself.

STRUCTURES

Reference the following structures, answer the following prompts

```
#define MAX 30;
typedef char String[50];

typedef struct{
  String bookName;
  String authorName;
  String publishingHouse;
  int publishingYear;
  int totalPageNum;
}Book;
```

```
typedef struct{
  Book elem;
  int next;
}BookNode;

typedef struct{
  BookNode books[MAX];
  int avail;
}BookShelf;
```

```
typedef struct{
   int first;
   int count;
}Genre;

Genre YA;
Genre Fantasy;
Genre Manga;
```

BookShelf is the heapspace. Assume it is a populated list and assume if traversal, to use the int pointer, `*trav`.

1. Display the bookName of the 1st book of the YA genre.
2. Update the publishing year of the last book of the Fantasy genre.
3. Interchange the first and last book of the Manga genre.

Coding:

Operations:

```
int allocSpace(BookShelf *B){
   int retVal=B->avail;

   if(retVal!=-1){
      B->avail =
         B->books[retVal].next;
   }

   return retVal;

}
```

```
void deallocSpace(BookShelf *B,int ndx){

   if(ndx!=-1 && ndx<MAX){
      B->books[ndx].next=B->avail;
      B->avail=ndx;
   }

   return retVal;

}
```

1. Given pointer to BookShelf BS and Genre YA, the function will return 0 if all the books in the top shelf are not unique. If they are, then the function will return 1.
    a. Function Header
    b. Function Call
    c. Code of the Function
    d. **Simulation**

2. Given pointer to BookShelf BS and Genre pointer Manga, the function will delete all occurrences of the specified publishingHouse.
    a. Function Header
    b. Function Call
    c. Code of the Function
    d. **Simulation**